Dynamic Model Inversion of a Soft Robot Using Servo-Constraints

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***Abstract*— Due to their flexibility, soft robots offer several advantages over their rigid counterparts in applications such as sensitive manipulation and human-machine interaction. For dynamic control, models must account for elastic deformation, underactuation, and redundancy. However, analytical inversion of forward dynamics is typically infeasible due to the complexity of the model.**

Beam models, such as the piecewise constant curvature (PCC) model, are commonly used when modeling the forward dynamics of soft robots. For control purposes, however, inverse models are required. In this work, we propose obtaining the inverse dynamic model of a soft robot from the forward dynamic PCC model using the servo-constraint approach. We analyze the system's internal stability and validate the model on a physical soft robot. Experimental results demonstrate the effectiveness of our approach in real-time dynamic trajectory tracking.

I. Introduction

Soft robotics is a growing subfield of robotics research. Due to their flexible nature, soft robots offer some advantages over their rigid counterparts and expand possibilities for new applications such as sensitive grasping, manipulation, and less constrained motion [1]. However, this flexibility comes at the cost of an infinite number of degrees of freedom, requiring new modeling and control schemes. The challenge is particularly pronounced in fast motions, where strong dynamic effects become significant.

In the literature, a distinction is made between quasi-static models and controllers, which neglect dynamics, and dynamic models and controllers. However, the terms quasi-static and kinematic are frequently used interchangeably. Most of the existing control algorithms for soft robots are based on purely quasi-static modeling [2], [3], [4], [5]. Such models allow for simple and accurate control, as long as the motions are slow enough for dynamics to be neglected. With emerging applications demanding faster and more precise motions, dynamic models and controllers are increasingly required and are currently being developed for soft robots.

Real-time dynamic control remains challenging due to the infinite dimensionality of the robot's state space [6]. Differentiation of deeply nested time-dependent variables in the kinematics leads to highly complex equations that are computationally expensive to solve in real time [7], [2]. Moreover, feedback control requires integrating sensors into the soft robot, which is difficult because the sensors must withstand the typically high strains without compromising softness. To overcome this, external camera tracking systems are often used, but their reliance on a continuous line of sight limits applicability across different environments. Feedforward control, alone or in combination with simple feedback, can greatly reduce the need for sensor measurements, enabling simpler, more robust, and less expensive soft robot designs. Therefore, accurate dynamic forward control of soft roots remains an important and still largely unsolved problem.

Many works have addressed the modeling of forward dynamics [7], [8], [9], [10]. However, precise output tracking requires determining system inputs that achieve a desired output, which is provided by an inverse model. Analytical model inversion is typically challenging due to the strong nonlinearities and complexities of soft robots. Furthermore, soft robots are controlled with a limited amount of actuators and are therefore typically highly underactuated, meaning not all degrees of freedom can be independently controlled. This further complicates model inversion. To address this, approaches from related fields can be considered. One such method is the addition of servo-constraints [11], a model inversion technique widely applied to complex underactuated multibody systems [12], [13].

A. Contribution

In this work, we present open-loop dynamic trajectory tracking control for a soft robot. First, we introduce our real-world test system. Secondly, we derive the forward dynamics from a kinematic piecewise constant curvature (PCC) model using the Newton–Euler formalism. Afterwards, we invert the model using the servo-constraints approach and analyze internal stability. Finally, we experimentally identify the remaining unknowns of our self-built robot and validate the effectiveness of the approach through tracking dynamic trajectories with the robot's tip.

II. Soft Robot Test System

Fig. 1 shows the beam-shaped soft robot used in this work. The robot's body is made from "HT45" silicone in a molding process. It is stiff along the longitudinal axis but allows bending in the x- and y-directions, so that the tip moves on a hemispherical surface around the robot. The robot is redundantly actuated by $N_{\text{tendons}} = 3$ tendons, placed symmetrically around the circumference and running along the outside of the body. Six notches are embedded along the body to guide deformation. Three servos are used to directly control the length s_j of the tendons $j \in 1 \dots N_{\text{tendons}}$. For the derivation of the dynamic model, we first consider the forces $\boldsymbol{u} = \begin{bmatrix} F_1 & F_2 & F_3 \end{bmatrix}^\top$ induced by the tendons as the system input. Later, during control,

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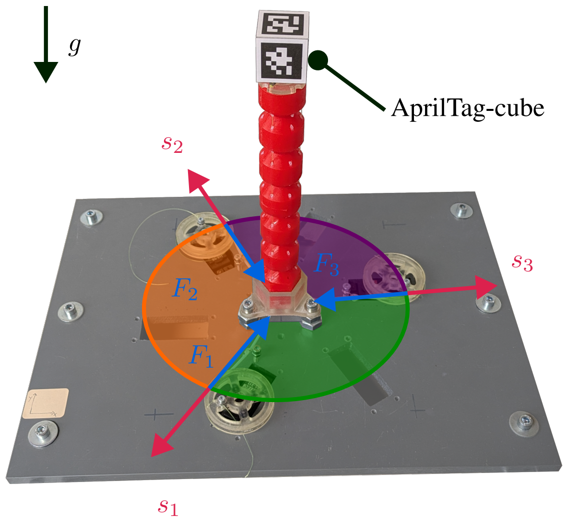

Fig. 1: Beam-shaped tendon-actuated soft robot test system: The tendons s_j are guided through holes in the soft body and pulled by the three motors $j = 1 \ldots 3$. For reference measurements, a down-facing camera tracks the Apriltag-cube which is attached to the soft robot (red body).

the tendon lengths s_j are used as input instead. Because of the redundant actuation, only two of the three tendon lengths s_j can be selected independently. Otherwise, the tendons would become loose or break. As system output, we use the projection $\boldsymbol{z} \in \mathbb{R}^2$ of the tip position onto the x-y-plane. For reference measurements, we use an external camera tracking system based on AprilTags [14]. A cube of size $l_{\text{cube}} = 0.03\,\text{m}$ with AprilTags from the *36h11* family attached to its sides and top is mounted at the tip of the robot. A complete list of robot parameters is provided in Table I.

TABLE I: Characteristics of the soft robot test system.

Dimension	Symbol	Unit	Value
Robot length	l_{robot}	m	0.19
Robot radius	r_{robot}	m	0.015
Length of tendon guide holes	l_{guide}	m	0.014
Circle radius of holes	r_{guide}	m	0.0125
End-effector cube mass	m_{cube}	kg	0.017
Mass of silicone body	m_{body}	kg	0.117
Young's modulus	E	MPA	2.04
Shore-hardness (HT 45-silicone)	H	ShA	45
Number of actuation tendons	N_{tendons}	-	3

III. Forward Model of the Soft Robot

In this section, the derivation of the dynamic forward model using the PCC model is briefly described. We discuss actuation redundancy and introduce additional model parameters that influence control.

A. Piecewise Constant Curvature Model

Approaches for modeling soft robot kinematics are commonly based on finite element (FE)-techniques, beam models, such as the Cosserat rod [15], or data-driven models [16], [17]. The PCC model is one of the simplest and most intuitive models for beams undergoing large deformations.

1) Discretization: In PCC, a beam is discretized into $i = 1 \ldots N_{\text{pieces}}$ pieces of constant curvature using a 1D FE mesh. Each piece consists of a massless elastic link, at the end of which a mass-loaded disk is mounted. The stiffness and damping properties are aggregated in the elastic link, and the inertia properties are aggregated in the disk. Bending in two spatial directions, torsion and strain can be modeled with the PCC model. Shear is neglected. The elongation mode is usually very stiff compared to the bending and torsion modes. Therefore, it is often neglected, as is the case in the following. Torsion is also neglected, which is often the case, since in many applications there are no or only very small torsional moments.

In this work, we explore different configurations for $N_{\text{pieces}} \in [1, \ldots, 6]$. Each piece has length l_i and mass m_i. For the case $N_{\text{pieces}} = 6$, we set $m_{N_{\text{pieces}}} = \frac{m_i}{2} + m_{\text{cube}}$ for the last piece. We model each piece as a solid cylinder with the inertia

$$I_{xx} = I_{yy} = \frac{m_i}{12}(3r_{\text{robot}}^2 + l_i^2), \tag{1}$$

$$I_{zz} = \frac{m_i}{2} r_{\text{robot}}^2. \tag{2}$$

2) Parametrization: There are various parameterizations of the PCC model [2], most of which have problems with removable singularities in the straight configuration [18]. We use a parameterization based on [19] and treat the remaining singularities with a Taylor series expansion around the removable singularities.

Two rotations μ_i and ν_i describe the deformation of a piece, as shown in Fig. 2. The vector $\boldsymbol{\rho}$ points to the center of rotation

$$\boldsymbol{\rho}_{i,\text{loc}} = \frac{l_i}{\varphi_i^2} \begin{bmatrix} \nu_i & -\mu_i & 0 \end{bmatrix}^\top, \tag{3}$$

where $\varphi_i = \sqrt{\nu_i^2 + \mu_i^2}$ is the total bending angle.

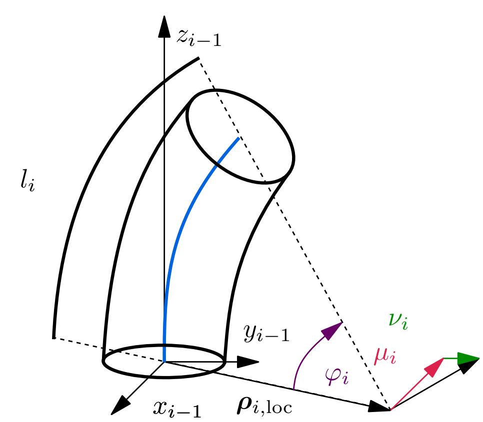

Fig. 2: Parameterization according to [19].

For each piece i, the rotation matrix $\boldsymbol{R}_{i-1,i}$ and position matrix $\boldsymbol{p}_{i,\text{loc}}$ describe the piece's tip (bottom frame of the adjacent piece) in the current bottom, piece-fixed frame

$$\boldsymbol{p}_{i,\text{loc}} = \begin{bmatrix}-\rho_{i,\text{loc},x}\sigma_i & -\rho_{i,\text{loc},y}\sigma_i & \|\boldsymbol{\rho}_{i,\text{loc}}\|S_{\varphi_i}\end{bmatrix}^\top, \quad (4)$$

$$\boldsymbol{R}_{i-1,i} = \begin{bmatrix}\sigma_i\bar{\nu}_i^2+1 & \sigma_i\bar{\mu}_i\bar{\nu}_i & \bar{\nu}_i\sin\varphi_i \\ \sigma_i\bar{\mu}_i\bar{\nu}_i & \sigma_i\bar{\mu}_i^2+1 & \bar{\mu}_i\sin\varphi_i \\ -\bar{\nu}_i\sin\varphi_i & -\bar{\mu}_i\sin\varphi_i & \cos\varphi_i\end{bmatrix}, \quad (5)$$

where $\sigma_i = \cos(\varphi_i) - 1$, $\bar{\nu}_i = \frac{\nu_i}{\varphi_i}$ and $\bar{\mu}_i = \frac{\mu_i}{\varphi_i}$, and $\rho_{i,\text{loc},x}$, and $\rho_{i,\text{loc},y}$ are the x and y entries of $\boldsymbol{\rho}_{i,\text{loc}}$. With the global frame placed at x_0-y_0, we can transform $\boldsymbol{p}_{i,\text{loc}}$ and $\boldsymbol{R}_{i-1,i}$ for each piece to the global frame with

$$\boldsymbol{R}_i = \begin{cases}\boldsymbol{R}_{i,\text{loc}} & \text{for} \quad i=1 \\ \boldsymbol{R}_{i-1}\cdot\boldsymbol{R}_{i-1,i} & \text{for} \quad i>1\end{cases}, \quad (6)$$

$$\boldsymbol{p}_i = \begin{cases}\boldsymbol{p}_{i,\text{loc}} & \text{for} \quad i=1 \\ \boldsymbol{p}_{i-1}+\boldsymbol{R}_{i-1}\cdot\boldsymbol{p}_{i,\text{loc}} & \text{for} \quad i>1\end{cases}. \quad (7)$$

B. Internal and External Loads

Internal moments result from the robot's bending stiffness. Although most soft robots have typically strong damping, we neglect resulting damping forces in the dynamic model, as experiments have shown a damping ratio of only $\delta \approx 0.0032$ for the test system used here. We only consider damping in the stability analysis in Sect. IV-B.2. The x-y-directional bending can be calculated by modeling the robot as a homogeneous beam [20]. The magnitude of the bending moment that acts on each piece is given by

$$l_{i,\text{bend}} = EJ_{xx}\cdot\frac{\varphi_i}{l_i}, \quad (8)$$

where J_{xx} denotes the second moment of area around the x-axis. Considering the free-body diagram of a single elastic piece, each piece i experiences a bending moment at each end: one due to its own bending, $\boldsymbol{l}_{i,\text{bend}}$, and one induced by the adjacent piece, $\boldsymbol{l}_{i+1:i,\text{bend}}$, which are obtained by

$$\boldsymbol{l}_{i,\text{bend}} = -l_{i,\text{bend}}\cdot\boldsymbol{n}_{i,\text{bp}}, \quad (9)$$

$$\boldsymbol{l}_{i+1:i,\text{bend}} = l_{i+1,\text{bend}}\cdot\boldsymbol{n}_{i+1,\text{bp}}. \quad (10)$$

Here $\boldsymbol{n}_{i,\text{bp}}$ is the normal vector of the bending plane

$$\boldsymbol{n}_{i,\text{bp}} = \boldsymbol{R}_{i-1}\cdot\begin{bmatrix}-\sin(\theta_i) & \cos(\theta_i) & 0\end{bmatrix}^\top, \quad (11)$$

where $\theta_i = \arctan 2(\mu_i, \nu_i)$ denotes the rotation angle of the bending plane. Note that the combination of the trigonometric functions causes zero division for the robots straight pose and therefore requires Taylor approximations of sin and cos at this configuration.

As the robot is positioned upright, the only external force results from gravity in negative z-direction. The gravitational force of each piece in the global frame is

$$\boldsymbol{f}_{g,i} = \begin{bmatrix}0 & 0 & -m_i g\end{bmatrix}^\top. \quad (12)$$

C. Actuation Loads

Actuation of the robot is achieved via three tendons routed through holes in the robot's body. We neglect tendon friction, which results in constant tendon forces F_j along each tendons. The positions of the holes guiding the tendons are defined in the local frame as

$$\boldsymbol{r}^l_{i,j,\text{loc}} = \boldsymbol{R}^z(\phi_j)\begin{bmatrix}r_\text{guide} & 0 & -l_\text{guide}/2\end{bmatrix}^\top, \quad (13)$$

$$\boldsymbol{r}^u_{i,j,\text{loc}} = \boldsymbol{R}^z(\phi_j)\begin{bmatrix}r_\text{guide} & 0 & l_\text{guide}/2\end{bmatrix}^\top, \quad (14)$$

where $\phi_j \in \{0°, 120°, 240°\}$ for $j = 1\ldots 3$. As shown in Fig. 3, r_guide denotes the distance from the center axis to the guide holes, l_guide is the length of the holes, and $\boldsymbol{R}^z(.)$ represents the rotation matrix around the z-axis. Transformation into the global coordinate frame yields

$$\boldsymbol{r}^q_{i,j} = \boldsymbol{p}_i + \boldsymbol{R}_i\cdot\boldsymbol{r}^q_{i,j,\text{loc}}, \quad (15)$$

for $q \in \{u, l\}$. To determine the forces exerted by the tendons on the pieces of the soft robot, we cut the pieces and the tendons free, as shown in Fig. 4. Without a loss of generality, it can be assumed that the tendons are fixed in the tendon guides. Except for the first and last piece, each tendon exerts an incoming force $\boldsymbol{f}_{i,i-1,j,\text{tendon}}$ and an outgoing force $\boldsymbol{f}_{i,i+1,j,\text{tendon}}$ on each piece. The magnitudes of these forces are equal to the tendon forces F_j, while their direction can be determined from the orientation of the incoming and outgoing tendons. The effective force directions are obtained using the positional vectors of two pieces pointing towards adjacent guides. The normalized direction vectors for the forces pointing towards the preceding piece are

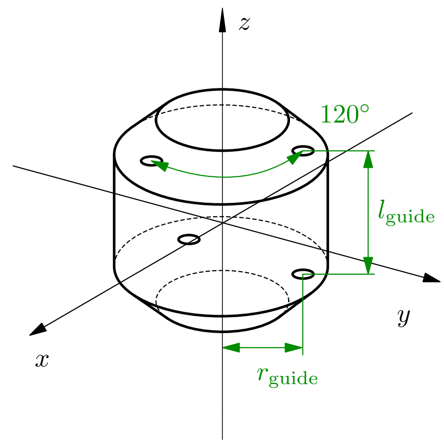

Fig. 3: Positions of the holes that guide the tendons.

$$\boldsymbol{c}_{i:i-1,j} = \frac{\boldsymbol{r}^u_{i-1,j}-\boldsymbol{r}^l_{i,j}}{\|\boldsymbol{r}^u_{i-1,j}-\boldsymbol{r}^l_{i,j}\|}, \quad (16)$$

and similar for the adjacent piece

$$\boldsymbol{c}_{i:i+1,j} = \frac{\boldsymbol{r}^l_{i+1,j}-\boldsymbol{r}^u_{i,j}}{\|\boldsymbol{r}^l_{i+1,j}-\boldsymbol{r}^u_{i,j}\|}. \quad (17)$$

Note that for the last piece $i = N_\text{pieces}$ we have $\boldsymbol{c}_{i:i+1,j} = \boldsymbol{0}$. Then, the actuation forces for each piece are given by

$$\boldsymbol{f}_{i:i-1,\text{tendon}} = \sum_{j=1}^{3}\boldsymbol{c}_{i:i-1,j}\cdot F_j, \quad (18)$$

$$\boldsymbol{f}_{i:i+1,\text{tendon}} = \sum_{j=1}^{3}\boldsymbol{c}_{i:i+1,j}\cdot F_j. \quad (19)$$

As a result of the eccentrically applied forces, each piece experiences moments during actuation. The moments that act on each piece are

$$\boldsymbol{l}_{i:i-1,\text{tendon}} = \sum_{j=1}^{3} \left(\boldsymbol{R}_i \cdot \boldsymbol{r}^l_{i,j}\right) \times \left(\boldsymbol{c}_{i:i-1,j} \cdot F_j\right), \tag{20}$$

$$\boldsymbol{l}_{i:i+1,\text{tendon}} = \sum_{j=1}^{3} \left(\boldsymbol{R}_i \cdot \boldsymbol{r}^u_{i,j}\right) \times \left(\boldsymbol{c}_{i:i+1,j} \cdot F_j\right). \tag{21}$$

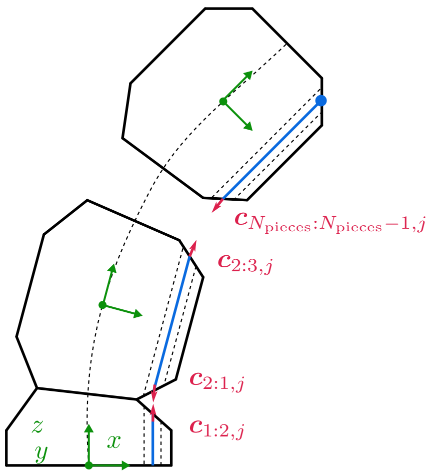

Fig. 4: Forces on each piece induced by tendon actuation.

D. Modeling Tendons

For application to the physical test system, a relationship between tendon forces $\boldsymbol{u}$ and tendon length $\boldsymbol{s}$ must be derived, since the input to the physical robot is the tendon length $\boldsymbol{s}$, whereas the tendon forces $\boldsymbol{u}$ are considered for modeling. Each tendon length s_j consists of a geometric change $s_{\text{geo},j}$ and an elastic change $s_{\text{stiff},j}$, accounting for both the elongation of the tendon itself and the robot's elongation under load

$$s_j = s_{\text{geo},j} + s_{\text{stiff},j}. \tag{22}$$

Both are displayed in Fig. 5. The geometric change in length of a single tendon is given by

$$s_{\text{geo},j} = l_{\text{cab},\boldsymbol{y}_0} - \sum_{n=0}^{N_\text{pieces}} \left(\|\boldsymbol{r}^l_{n+1,j} - \boldsymbol{r}^u_{n,j}\|\right)|_{\boldsymbol{y}}. \tag{23}$$

Modeling the elasticity of the tendons with a virtual spring of stiffness c_j provides a mapping from the tendon force F_j to its length s_i. In addition, variations in tendon mounting and pretension must be considered. To account for these uncertainties, a deformation coefficient b_j is introduced for $s_{\text{geo},j}$, yielding

$$s_j = b_j s_{\text{geo},j} + \frac{F_j}{c_j}. \tag{24}$$

The robot's stiffness along its longitudinal direction introduces redundant actuation, which requires an input mapping. The robot's workspace is divided into three segments along the tendon guides, as can be seen in Fig. 1. The two forces spanning the actuation region are set as positive, while the force outside of this sub-workspace is set to zero. This approach results in smaller forces compared to alternative methods, in which all forces are generally higher. The distribution of the workspace is characterized by an angle ψ defined in the x-y plane, which provides a mapping

$$\boldsymbol{u} = \boldsymbol{\Phi}^\top \cdot \boldsymbol{u}_\text{red}, \tag{25}$$

where $\boldsymbol{u}_\text{red}$ denotes the reduced unknown input, consisting of the two active forces. The mapping matrix $\boldsymbol{\Phi}$ depends on the location of the desired tip position z_y^des and z_x^des, and is given by

$$\boldsymbol{\Phi} = \begin{cases} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} & \text{for} \quad \psi \in [0, 120°) \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \text{for} \quad \psi \in [120°, 240°) \\ \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} & \text{for} \quad \psi \in [240°, 360°) \end{cases} \tag{26}$$

with $\psi = \arctan 2(z_y^\text{des}, z_x^\text{des})$.

E. Equations of Motion

With $\boldsymbol{y}_i = \begin{bmatrix}\mu_i & \nu_i\end{bmatrix}^\top$, we define the generalized coordinates $\boldsymbol{y} = \begin{bmatrix}\boldsymbol{y}_1^\top & \dots & \boldsymbol{y}_{N_\text{pieces}}^\top\end{bmatrix}^\top$. The equations of motions are derived following the Newton-Euler formalism, and can be written as second-order ordinary differential equations (ODE) in the form

$$\boldsymbol{M}(\boldsymbol{y}, t)\ddot{\boldsymbol{y}} + \boldsymbol{k}(\boldsymbol{y}, \dot{\boldsymbol{y}}, t) = \boldsymbol{q}(\boldsymbol{y}, \dot{\boldsymbol{y}}, t) + \boldsymbol{B}(\boldsymbol{y})\boldsymbol{u}. \tag{27}$$

With $f = 2N_\text{pieces}$, $\boldsymbol{M} \in \mathbb{R}^{f \times f}$ denotes the mass matrix, $\boldsymbol{k} \in \mathbb{R}^f$ represents the Coriolis, centrifugal and gyroscopic forces, $\boldsymbol{q} \in \mathbb{R}^f$ are the applied forces, and $\boldsymbol{B} \in \mathbb{R}^{f \times N_\text{tendons}}$ is the input distribution matrix. The system output is the position of the tip of the soft robot $\boldsymbol{z}$.

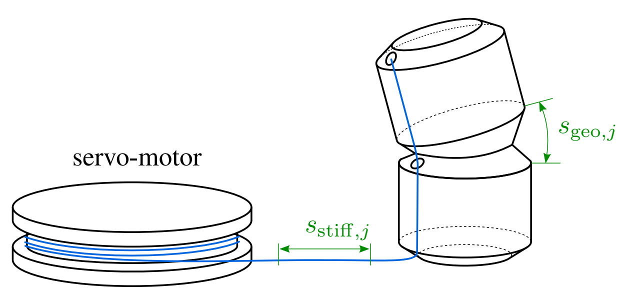

Fig. 5: The motors rotation causes the robot to bend and tendons to stretch.

IV. Model Inversion using Servo-Constraints

For feedforward control, the inverse of the system dynamics given in Eq. (27) is required. We apply the servo-constraints approach [21] to obtain the inverse model, a common method for controlling underactuated multibody systems [22], [23].

A. Inverse Model

To obtain the inverse dynamics of the system, the equations of motion (27) are extended with additional servo-constraints that force the system output $\boldsymbol{z}$ to follow the desired trajectory $\boldsymbol{z}^{\text{des}}$. This can be formulated as a set of differential algebraic equations (DAE)

$$\begin{aligned} \boldsymbol{v} &= \dot{\boldsymbol{y}}, \\ \boldsymbol{M}(\boldsymbol{y},t)\dot{\boldsymbol{v}} &= \boldsymbol{q}(\boldsymbol{y},\boldsymbol{v},t) - \boldsymbol{k}(\boldsymbol{y},\boldsymbol{v},t) + \boldsymbol{B}(\boldsymbol{y})\boldsymbol{u}, \\ \boldsymbol{c}(\boldsymbol{y},t) &= \boldsymbol{z}(\boldsymbol{y}) - \boldsymbol{z}^{\text{des}}(t) = \boldsymbol{0}. \end{aligned} \tag{28}$$

This inverse system is fully described and can be solved for the state vector $\boldsymbol{x} = \begin{bmatrix}\boldsymbol{y}^\top & \boldsymbol{v}^\top & \boldsymbol{u}^\top\end{bmatrix}^\top$.

B. Internal Dynamics and Invertibility Analysis

The DAE in Eq. (28) can only be solved by forward time integration if the inverse system is stable. To ensure this, we analyze the system's invertibility and internal dynamics. In the following analysis, we only consider the case $N_{\text{pieces}} = 2$, which provides a multibody configuration while keeping the computational cost low when solving Eq. (28) using an implicit Euler approach. Numerical experiments have shown that the solver remains stable for $N_{\text{pieces}} \geq 2$, and therefore we assume that stability is maintained for finer discretizations.

1) Differentiation Index: We consider the differentiation index, which quantifies the number of times the algebraic system constraints must be differentiated before the DAE can be transformed into ODE form [24]. The differentiation index is important numerically, as many solvers can only handle systems of index 1. In the present system, the input $\boldsymbol{u}$ appears after differentiating $\boldsymbol{c}(\boldsymbol{y},t)$ twice, resulting in a differentiation index of 3, which is typical for multibody systems [25]. Therefore, we employ the implicit Euler algorithm, which can handle index 3 systems and enables real-time application. Substituting Eq. (27) for $\dot{\boldsymbol{v}}$ then provides the feedforward control input

$$\widetilde{\boldsymbol{u}}(t) = (\widetilde{\boldsymbol{H}}\widetilde{\boldsymbol{M}}^{-1}\widetilde{\boldsymbol{B}})^{-1}(\ddot{\boldsymbol{z}}^{\text{des}} - \widetilde{\boldsymbol{H}}\widetilde{\boldsymbol{M}}^{-1}(\widetilde{\boldsymbol{q}} - \widetilde{\boldsymbol{k}}) - \widetilde{\boldsymbol{H}}\boldsymbol{v}). \tag{29}$$

Here, $\boldsymbol{H} = \frac{\partial \boldsymbol{h}}{\partial \boldsymbol{y}}$ and the tilde denotes quantities evaluated at $\widetilde{\boldsymbol{x}}$, the state required for the tip to reach $\boldsymbol{z}^{\text{des}}$. For full rank of the coupling matrix $\mathbf{Y}(\boldsymbol{y}) = \widetilde{\boldsymbol{H}}\widetilde{\boldsymbol{M}}^{-1}\widetilde{\boldsymbol{B}}$, the input $\boldsymbol{u}$ directly influences all system outputs [21]. We find that $\operatorname{rank}(\mathbf{Y}) = 2$, which is equal to the number of unknown system inputs. Thus, a control law can be established straightforwardly.

A related important property from a control-theoretic perspective is the relative degree r_{rel}, which specifies the number of times the desired output trajectory $\boldsymbol{z}^{\text{des}}$ must be differentiated with respect to time to reveal the influence of the input [26]. From Eq. (29), it is evident that the desired trajectory $\boldsymbol{z}^{\text{des}}$ must be defined at the acceleration level, and hence $r_{\text{rel}} = 2$. For single-input-single-output systems, the differentiation index is shown to be equal to $r_{\text{rel}} + 1$ when the internal dynamics can be described by ODE [27], which typically holds for multibody systems [13].

2) Stability Analysis of the Inverse System: Full-specified motion is only given for a single-piece model $N_{\text{pieces}} = 1$, where $N_{\text{tendons}} = \dim(\boldsymbol{z}) = f = 2$. For finer discretization $N_{\text{pieces}} \geq 2$, internal dynamics may cause destabilization and servo-constraints must be designed through dynamic couplings in the system instead [21]. A nonlinear system is of minimum phase if it has stable zero dynamics [28]. Hence, we linearize the inverse model of Eq. (27) at a stationary point $\boldsymbol{x}_{\text{eq}}$. For small values around the point of equilibrium $\widetilde{\boldsymbol{y}} = \boldsymbol{y} - \boldsymbol{y}_{\text{eq}}$, $\widetilde{\boldsymbol{v}} = \boldsymbol{v} - \boldsymbol{v}_{\text{eq}}$ and $\widetilde{\boldsymbol{u}} = \boldsymbol{u} - \boldsymbol{u}_{\text{eq}}$ the linearized equations of motion are given by

$$\boldsymbol{M}_{\text{lin}}\dot{\widetilde{\boldsymbol{v}}} + \boldsymbol{D}_{\text{lin}}\widetilde{\boldsymbol{v}} + \boldsymbol{K}_{\text{lin}}\widetilde{\boldsymbol{y}} = \boldsymbol{B}_{\text{lin}}\widetilde{\boldsymbol{u}}, \tag{30}$$

$$\boldsymbol{H}_{\text{lin}}\widetilde{\boldsymbol{y}} = \boldsymbol{0}, \tag{31}$$

with $\boldsymbol{M}_{\text{lin}} = \boldsymbol{M}(\boldsymbol{y}_{\text{eq}})$, $\boldsymbol{B}_{\text{lin}} = \boldsymbol{B}(\boldsymbol{y}_{\text{eq}})$, $\boldsymbol{H}_{\text{lin}} = \boldsymbol{H}(\boldsymbol{y}_{\text{eq}})$,

$$\boldsymbol{D}_{\text{lin}} = \boldsymbol{C} + \left(\frac{\partial \boldsymbol{k}}{\partial \boldsymbol{v}} - \frac{\partial \boldsymbol{q}}{\partial \boldsymbol{v}}\right)\bigg|_{\boldsymbol{y}_{\text{eq}},\boldsymbol{v}_{\text{eq}}}, \tag{32}$$

$$\boldsymbol{K}_{\text{lin}} = \left(\frac{\partial \boldsymbol{k}}{\partial \boldsymbol{y}} - \frac{\partial \boldsymbol{q}}{\partial \boldsymbol{y}} - \frac{\partial \boldsymbol{B}\boldsymbol{u}_{\text{eq}}}{\partial \boldsymbol{y}}\right)\bigg|_{\boldsymbol{y}_{\text{eq}},\dot{\boldsymbol{y}}_{\text{eq}}}. \tag{33}$$

The linearized equations can be rewritten as

$$\boldsymbol{E}^*\dot{\widetilde{\boldsymbol{x}}} = \boldsymbol{A}^*\widetilde{\boldsymbol{x}}, \tag{34}$$

with the augmented state vector

$$\widetilde{\boldsymbol{x}} = \begin{bmatrix}\widetilde{\boldsymbol{y}}^\top & \widetilde{\boldsymbol{v}}^\top & \widetilde{\boldsymbol{u}}^\top\end{bmatrix}^\top, \tag{35}$$

and the two matrices

$$\boldsymbol{E}^* = \begin{bmatrix}\mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{M}_{\text{lin}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix}, \tag{36}$$

$$\boldsymbol{A}^* = \begin{bmatrix}\mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\boldsymbol{K}_{\text{lin}} & -\boldsymbol{D}_{\text{lin}} & \boldsymbol{B}_{\text{lin}} \\ \boldsymbol{H}_{\text{lin}} & \mathbf{0} & \mathbf{0}\end{bmatrix}. \tag{37}$$

Rearranging Eq. (34) gives the eigenvalue problem

$$(\boldsymbol{A}^* - \lambda\boldsymbol{E}^*)\mathbf{c}^* = \mathbf{0}. \tag{38}$$

Infinite eigenvalues result from algebraic constraints with infinitely fast motion [29]. Invariant poles $\lambda < \infty$ determine whether the system is minimum phase. Evaluating Eq. (27) at a point of equilibrium with $\boldsymbol{v} = \mathbf{0}$ and $\dot{\boldsymbol{v}} = \mathbf{0}$ for an arbitrary force $\boldsymbol{u} = \begin{bmatrix}1 & 1 & 0\end{bmatrix}^T$, and solving Eq. (38) for λ demonstrates the internal stability of the inverse system, as shown in Fig. 6. All λ lie in the left-hand plane, confirming that the system is minimum phase.

V. Experimental Evaluation

We study the effectiveness of this inversion approach on circular and triangular trajectories. First, we experimentally identify function approximations for the deformation coefficients b_j and spring stiffness coefficients c_j. Second, we evaluate the effects of the desired trajectory speed $v_{\max}$ and the discretization degree N_{pieces} on positional accuracy in trajectory tracking.

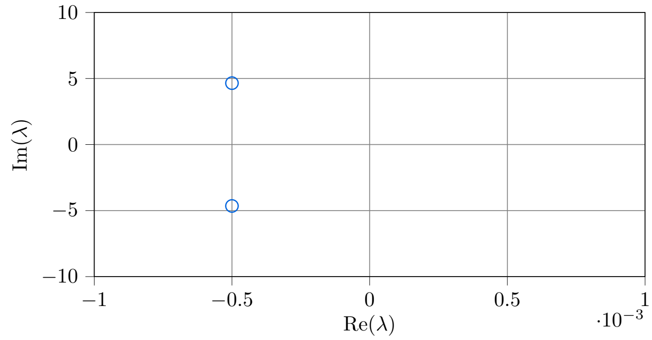

Fig. 6: Eigenvalues of the inverse system with 2 segments.

A. Actuation Coefficients Identification

First, we tune the deformation and spring stiffness coefficients b_j and c_j. For this purpose, the average positional error of the end-effector is introduced as

$$e_{\text{avg}} = \text{mean}\left(\|\boldsymbol{z}_t^{\text{des}} - \boldsymbol{z}_t^{\text{meas}}\|\right), \tag{39}$$

where $\boldsymbol{z}_t^{\text{meas}}$ and $\boldsymbol{z}_t^{\text{des}}$ denote the measured and desired position of the end-effector in the x-y plane, respectively. The coefficients are determined using circular trajectories centered at the robot's base. As in the stability analysis, the DAE in Eq. (28) are solved using implicit Euler method. We tested varying step sizes $\Delta t \in \{0.01\,\text{s}, 0.02\,\text{s}, 0.1\,\text{s}\}$, but found no significant effect on performance. Therefore, we use $\Delta t = 0.01\,\text{s}$ in the following. Early experiments showed that constant b_j and c_j, as well as r_c-based linear scaling, are insufficient for precise tracking. Consequently, we tune the coefficients for a selection of circular trajectories experimentally and fit second-order polynomial functions dependent on the current distance between the robot's tip and its center. Larger radii are chosen for sampling, since for smaller radii, the actuation forces and, consequently, bending are small. The resulting functions $c_j(\|\boldsymbol{z}_t^{\text{des}}\|)$ and $b_j(\|\boldsymbol{z}_t^{\text{des}}\|)$ are shown in Fig. 7.

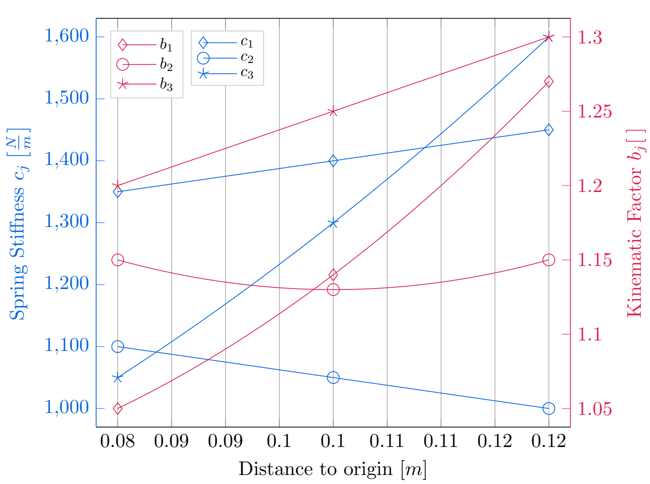

Fig. 7: Polynomials to approximate the coefficients b_j and c_j and measured sample points at $r_c \in \{0.08\,\text{m}, 0.1\,\text{m}, 0.12\,\text{m}\}$.

B. Trajectory Tracking

We introduce two distinctive trajectories to reveal the robot's dynamic behavior: a circular path and an equilateral triangle. Circular motion around the robot's base induces centrifugal effects, while the triangle path provides straight segments and sharp turns, challenging the actuation and enabling exploration of the full range of the approximator functions. In both cases, the robot first moves out of the straight configuration before tracking the desired trajectories.

Under the imposed twice-differentiability constraint, both trajectories follow the velocity profile shown in Fig. 8. The trajectories reach constant acceleration and deceleration of $a_{\text{max}} = 0.1\,\frac{\text{m}}{\text{s}^2}$ with a maximum jerk $j_{\text{max}} = 0.1\,\frac{\text{m}}{\text{s}^3}$. Throughout the full time period, it is ensured that the robot completes at least one full lap at v_{max}. During the initial deflection, where the same velocity profile was applied, the solver occasionally struggled to find solutions for certain configurations. To overcome this, the initial position was slightly adjusted. This issue can be attributed to limitations of the PCC parameterization in the straight configuration, as discussed in detail in [18].

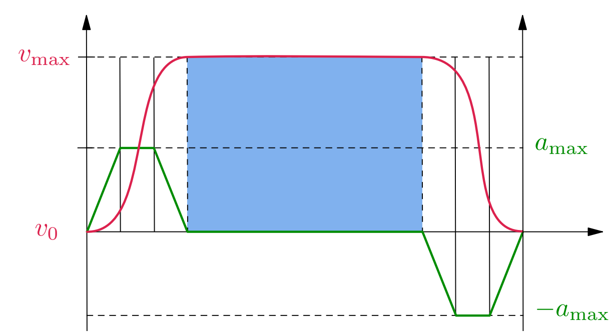

Fig. 8: Within the constant velocity phase of v_{max} (blue), the robot takes at least one full lap.

1) Circular Trajectory: Three circles with radii of $4\,\text{cm}$, $8\,\text{cm}$, and $12\,\text{cm}$ are studied, with the largest approaching the maximum workspace dimension.

The average positional error e_{avg} for varying v_{max} and r_c and $N_{\text{pieces}} = 6$ is shown in Fig. 9. The results indicate that e_{avg} increases with v_{max} for each circle radius. In addition, e_{avg} is larger for circles of greater size, where the actuation forces are higher. It should be noted that the measurements are affected by noise, as the camera occasionally failed to detect the AprilTags due to vibrations of the robot.

Although real-time control is achieved for $N_{\text{pieces}} = 6$ kinematics, the offline symbolic derivation of the equations of motion (27) at this high discretization is computationally intensive. For comparison, derivation of the dynamic model requires around $2\,\text{min}$ for $N_{\text{pieces}} = 1$ and up to a full week for $N_{\text{pieces}} = 6$ on a high-performance workstation (Intel Xeon W-2245, 64 GB RAM). This raises the question whether a less complex representation affects dynamic tracking quality. Fig. 10 shows the results for $N_{\text{pieces}} \in \{1, 3, 6\}$. A clear decrease in accuracy is observed for simpler models, where strong deformations are captured less precisely.

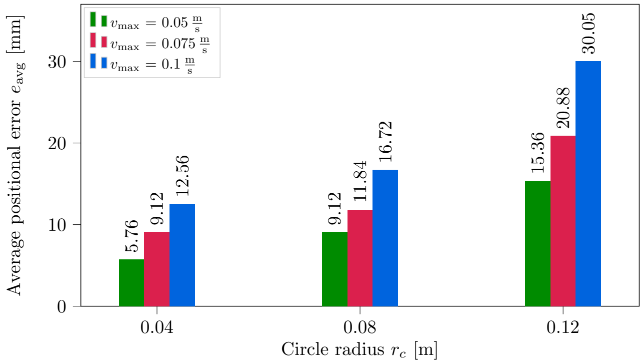

Fig. 9: Error e_{avg} of tracking circles of varying radii r_c and $v_{\max}$.

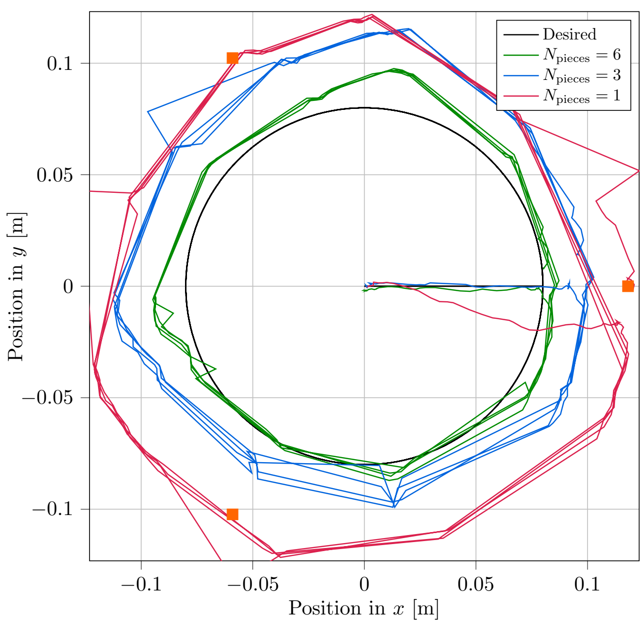

Fig. 10: Tracking a circle of $r_c = 0.08\,\text{m}$ with $v_{\max} = 0.1\,\frac{\text{m}}{\text{s}}$ for varying $N_{\text{pieces}} \in \{1, 3, 6\}$. The orange squares mark the servo-motor locations.

2) Triangular Trajectory: We examine two triangular trajectories with rounded corners to ensure differentiability: one with corners aligned to the actuator positions (aligned) and another mirrored vertically (mirrored). We set $v_{\max} = 0.1\,\frac{\text{m}}{\text{s}}$ and $N_{\text{pieces}} = 6$.

Fig. 11 shows the desired and measured tip positions. In both cases, the robot is capable of accurately tracking the straight segments, with strong deformations occurring primarily near the corners. Especially for the aligned trajectory, the sharp turns at the corners are particularly noticeable.

3) Interpretation: Overall, the approach demonstrates strong tracking performance for dynamic motions. However, positional deviations are observed in all cases, which can be attributed to several factors. First, as mentioned earlier, measurements are affected by noise. Second, the tendons are attached only with strong tape, leading to loosening over time. Third, variations in the assembly and pretension of the tendons complicated the tuning of b_j and c_j. As a result, repetitive deviations were challenging to avoid as the function approximators scale only with $\|z_t^{\text{des}}\|$. In particular, the triangular paths reveal that the motor at $(0.12, 0)$ suffers from mounting inaccuracies, limiting the robot's ability to reach the far-left and far-right corners. This effect especially occurred near the workspace boundaries. Consequently, the robot overshoots at the opposite straight edges.

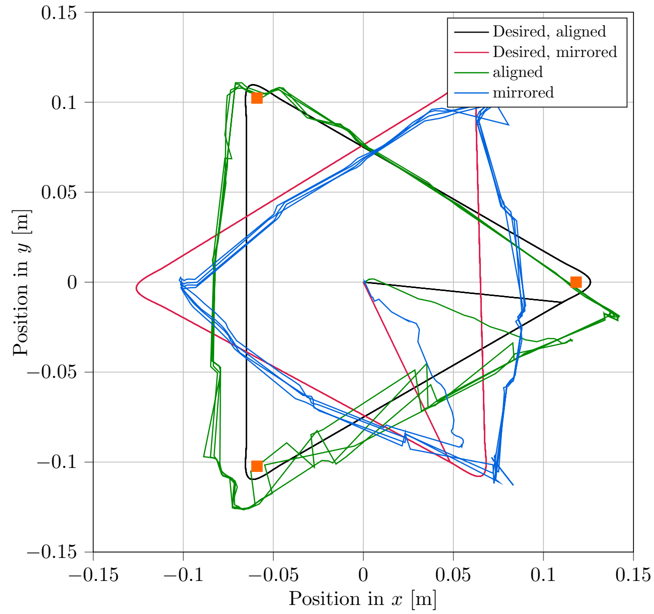

Fig. 11: Tracking equilateral triangles with $v_{\max} = 0.1\,\frac{\text{m}}{\text{s}}$ and $N_{\text{pieces}} = 6$. The orange squares mark the servo-motor locations.

VI. Conclusion

This work presents a model-based, feedforward, dynamic control approach for beam-shaped soft robots. For forward modeling, a dynamic piecewise constant curvature model is used, which is then inverted via the servo-constraints approach. The stability of the inverse model's internal dynamics is demonstrated. Trajectory tracking experiments were performed on a physical, beam-shaped, tendon-actuated soft robot to demonstrate the applicability of the proposed approach. While the method enables real-time control and achieves overall accurate trajectory tracking, the results reveal limitations in the actuation functions and sensitivity to motor limits, pretension, and mounting variations.

In future work, we plan to improve the system setup by using fixed tendons with constant pretension. We also aim to enhance the actuation functions using learning-based approaches to account for uncertainties. Lastly, we will extend this approach to more complex soft robotic systems.

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